



# higher education & training

Department:  
Higher Education and Training  
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTROTECHNOLOGY  
NQF LEVEL 2

(12040212)

21 November 2025 (Y-paper)  
13:00–16:00

Calculators may be used.

This question paper consists of 8 pages, 1 addendum and 1 formula sheet.

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**Search for NCV Exam Papers**

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**DEPARTMENT OF HIGHER EDUCATION AND TRAINING**  
**REPUBLIC OF SOUTH AFRICA**  
NATIONAL CERTIFICATE (VOCATIONAL)  
ELECTROTECHNOLOGY  
NQF LEVEL 2  
TIME: 3 HOURS  
MARKS: 100

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**INSTRUCTIONS AND INFORMATION**

1. Answer all the questions.
  2. Read all the questions carefully.
  3. Number the answers according to the numbering system used in this question paper.
  4. Use only a black or blue pen.
  5. Write neatly and legibly.
-

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-3-

### QUESTION 1

1.1 Indicate whether the following statements are TRUE or FALSE by writing only 'True' or 'False' next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1.1 In an atom, the number of electrons is equal to the number of protons.

1.1.2 The relationship between an electric current and a magnetic field is referred to as electromagnetism.

1.1.3 Copper is a good insulator.

1.1.4 Ohm's law states that current ( $I$ ) is inversely proportional to voltage ( $V$ ), provided that the resistance ( $R$ ) does not change.

1.1.5 The direction of the conventional flow of DC is from the positive to the negative terminal of the source.

(5 × 1) (5)

1.2 A mechatronic professional uses a 1-km long copper wire cable with a diameter of 6 mm. Calculate the resistance of the copper.

HINT: Take the resistivity of copper as  $1,72 \times 10^{-8} \Omega \cdot \text{m}$ . (3)

1.3 Study FIGURE 1 below and answer the questions that follow.

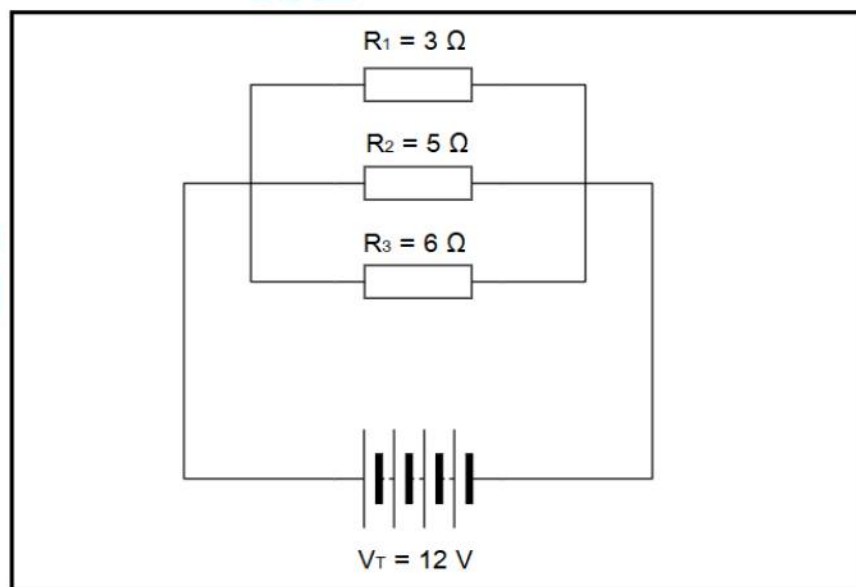


FIGURE 1

1.3.1 Refer to the way all the lamps are connected to each other and name the type of circuit connection. (1)

1.3.2 Calculate the total resistance of the circuit. (4)

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-4-

1.3.3 Calculate the total current flowing through the circuit. (2)

1.3.4 Calculate the total power drawn by the circuit. (2)

1.4 When sketching the lines of magnetic fields, there are some requirements that must be taken into consideration.

Explain THREE of these requirements.

(3)

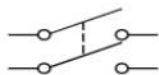
[20]

## QUESTION 2

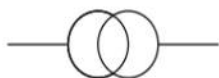
2.1 IEC symbols are used to represent wiring, components and apparatus in circuit diagrams.

Identify the following IEC symbols. Write only the answer next to the question number (2.1.1–2.1.5) in the ANSWER BOOK.

2.1.1



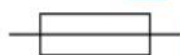
2.1.2



2.1.3



2.1.4



2.1.5



(5 × 1)

(5)

2.2 Draw a neat, labelled wiring diagram for a single luminaire controlled from two switches, both connected to the same circuit breaker. (6)

2.3 Electric cables are important for transporting electrical energy from power stations to various locations.

Name FOUR types of cables normally used in domestic installations.

(4)

12)

2.4

State the recommended circuit breaker size used in the domestic installation of the following sub-circuits:

2.4.1 Stove

2.4.2 Socket outlet

2.4.3 Geyser

2.4.4 Bell circuit

(4 × 1) (4)

2.5

In an electrical distribution board, two types of bars are used for safety and connectivity.

Name these TWO bars that are found in a DB box.

(2)

2.6

Define *skirting*.

(1)

2.7

The old saying of 'use the correct tool for the job' is very important and should always be adhered to.

Explain the uses of the following tools:

2.7.1 Side cutter

2.7.2 Spirit level

2.7.3 Bending spring

(3 × 1) (3)

[25]

### QUESTION 3

3.1

Analyse the block diagram in FIGURE 2 below and answer the questions that follow.

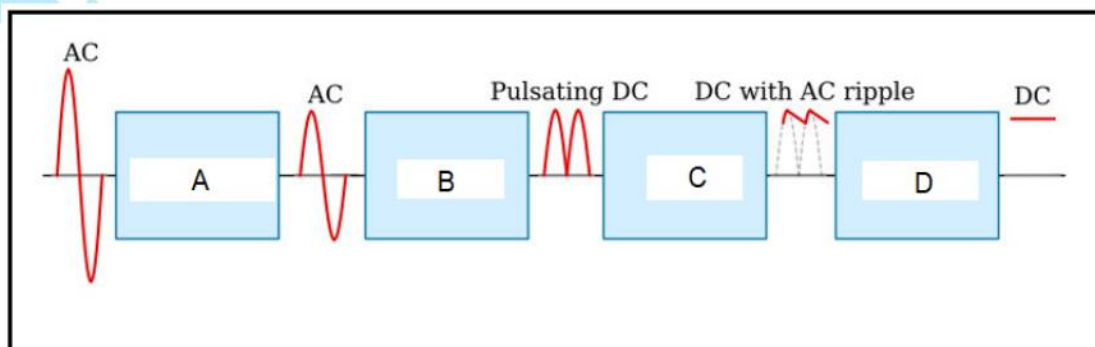


FIGURE 2

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- 3.1.1 Identify the name of the block diagram in FIGURE 2. (1)
- 3.1.2 Identify the components (A–D) shown in FIGURE 2. (4)
- 3.1.3 Explain the function of component C in the block diagram. (2)
- 3.2 Name THREE precautions when working with a switch-on soldering iron. (3)
- 3.3 Briefly explain how you would test a resistor in a circuit using a digital multimeter. (4)
- 3.4 It is important to tackle any task in the right sequence.  
Write FOUR important questions to consider when planning a task. (4)
- 3.5 A silicon-controlled rectifier (SCR) is a semiconductor device used for controlling power and rectifying AC into DC.  
Draw a neat, labelled diagram of the EIC symbol of an SCR. (3)
- 3.6 In an electronic workshop, there must be a toolkit that contains the basic hand tools needed to do any work.  
Name FOUR basic tools needed to replace components on a circuit board. (4)
- [25]**

#### QUESTION 4

- 4.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (4.1.1–4.1.5) in the ANSWER BOOK.
- 4.1.1 Which of the following is a common symptom of a short circuit in an electrotechnical circuit?
- A Circuit operates normally
  - B Circuit breaker trips
  - C Increased voltage
  - D Decreased current
- 4.1.2 What tool is commonly used to measure voltage in a circuit?
- A Ammeter
  - B Voltmeter
  - C Oscilloscope
  - D Multimeter

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4.1.3 In an electrical circuit, what does a blown fuse typically indicate?

- A Normal operation
- B Overcurrent condition
- C Short circuit
- D Both B and C

4.1.4 What is the primary purpose of a Megger tester?

- A To measure current
- B To check the integrity of the insulation
- C To measure voltage drop
- D To test continuity

4.1.5 When diagnosing a fault in a circuit, which of the following should be checked first?

- A Load devices
- B Power supply
- C Wiring connections
- D Circuit breakers

(5 × 1)

(5)

4.2 Mr Brown, a maintenance technician at ABC TVET College, is tasked with performing routine maintenance on a three-phase industrial motor used in the workshop.

List FIVE items that should be checked as part of the maintenance procedure.

(5)

4.3 Study the circuit in FIGURE 3 below and answer the questions that follow.

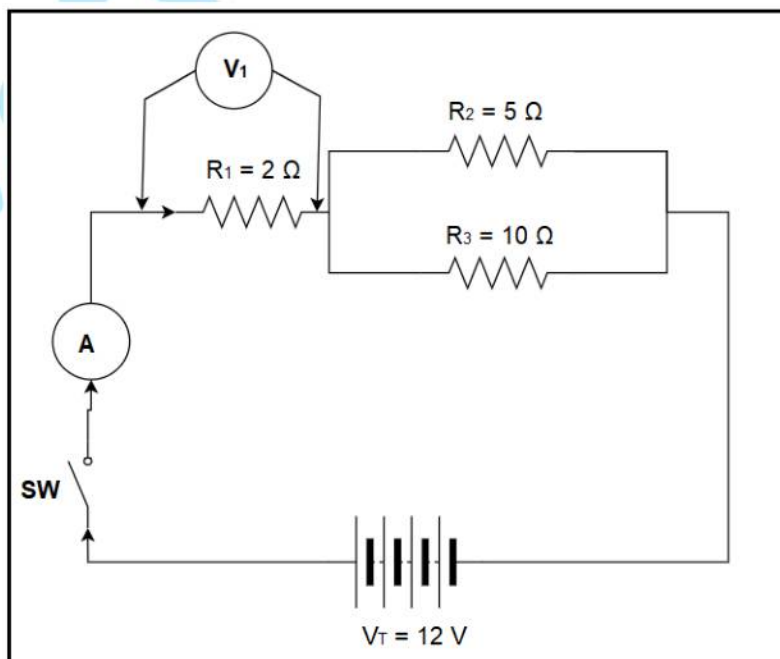


FIGURE 3

12)

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- 4.3.1 If the switch (SW) in FIGURE 3 is closed, what will the reading be on the ammeter (A)? (4)
- 4.3.2 What will the ammeter (A) reading be if  $R_2$  is experiencing an open-circuit fault? (2)
- 4.3.3 What will the voltmeter reading ( $V_1$ ) be if  $R_2$  is still experiencing an open-circuit fault? (2)
- 4.3.4 What will happen to the total resistance and current if resistor  $R_3$  is removed from the circuit? (2)
- [20]**

### QUESTION 5

Mr Mokoena of the cafeteria of CSG Factory had a stove fitted into the kitchen. He phoned on 2 April 2025 from his mobile phone, 072 345 6789, and requested an electrician to come out and identify a fault. His diagnosis was that when he touched the stove, he felt a little shock.

You arrive at the cafeteria at 09:20 and you first check the complete circuit. You find that there is no earth continuity from the stove to the DB. You repair the fault and leave after two hours.

Your supervisor, Mr Dlamini, has requested you to complete the maintenance record indicating all the work that you have done.

Refer to ADDENDUM A (attached) and complete the report. Remove ADDENDUM A from the question paper and place it in your ANSWER BOOK.

**[10]**

**TOTAL: 100**

12)

**ADDENDUM A**

**EXAMINATION NUMBER:**

|  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
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|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|

**QUESTION 5**

|                              |                    |             |
|------------------------------|--------------------|-------------|
| <b>ABC ELECTRIC</b>          |                    |             |
| 225 Botha Ave, Amp Park 2025 |                    |             |
|                              |                    | <b>DATE</b> |
| <b>Customer department</b>   |                    |             |
| <b>Customer phone</b>        |                    |             |
| <b>Work request</b>          |                    |             |
|                              |                    |             |
|                              |                    |             |
| <b>Work done</b>             |                    |             |
|                              |                    |             |
|                              |                    |             |
| <b>Remarks</b>               |                    |             |
|                              |                    |             |
|                              |                    |             |
|                              |                    |             |
| <b>Time of arrival</b>       |                    |             |
| <b>Time of departure</b>     |                    |             |
| <b>Work done by</b>          |                    |             |
|                              | <b>SUPERVISOR:</b> |             |

[10]

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## FORMULA SHEET

### ELECTROTECHNOLOGY LEVEL 2

$$1. \quad P = V I \cos \phi$$

$$2. \quad Q = V I \sin \phi$$

$$3. \quad S = V I$$

$$4. \quad V = I R$$

$$5. \quad P = V I$$

$$6. \quad X_L = 2\pi f L$$

$$7. \quad X_C = \frac{1}{2\pi f C}$$

$$8. \quad Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$9. \quad V_T = \sqrt{V_R^2 + (V_L - V_C)^2}$$

$$10. \quad V_1 = \left( \frac{R_1}{R_t} \right) V_T$$

$$11. \quad V_L = I X_L$$

$$12. \quad I_T = \sqrt{I_R^2 + (I_L - I_C)^2}$$

$$13. \quad \tan \phi = \frac{I_L - I_C}{I_R}$$

$$14. \quad Z = \frac{V}{I}$$

$$15. \quad A = \frac{\pi d^2}{4}$$

$$16. \quad R_T = R_1 + R_2 + R_3 + \dots + R_n$$

$$17. \quad \frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$$

$$18. \quad C_T = C_1 + C_2 + C_3 + \dots + C_n$$

$$19. \quad \frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_n}$$

$$20. \quad L_T = L_1 + L_2 + L_3 + \dots + L_n$$

$$21. \quad \frac{1}{L_T} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \dots + \frac{1}{L_n}$$

$$22. \quad Q = C V$$

$$23. \quad R_t = R_0 (1 + \alpha t)$$

$$24. \quad \frac{V_1}{V_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1}$$

$$25. \quad I_1 = \left( \frac{R_2}{R_1 + R_2} \right) I_T$$

$$26. \quad V_C = I X_C$$

$$27. \quad R = \frac{p l}{A}$$

$$28. \quad \tan \phi = \frac{X_L - X_C}{R}$$

$$29. \quad E = I^2 R t$$

$$30. \quad n_s = \frac{f}{p}$$

$$31. \quad E = V + I r$$